CLCNet: A Curriculum-Guided Contrastive Learning Network With Spectral-Hierarchical Attention for Hyperspectral-LiDAR Image Classification

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***Abstract*—The scarcity of labeled samples has long constrained hyperspectral–light detection and ranging (LiDAR) fusion for land-cover classification, making contrastive learning a promising paradigm. However, existing methods still suffer from false-negative interference and insufficient modeling of hard negatives, which degrade representation quality. To address these challenges, we propose a curriculum-learning-based contrastive learning network (CLCNet), which enhances contrastive learning from two perspectives. First, a spectral–hierarchical multihead attention (MHA) mechanism reorders and groups spectral channels to decouple attention across groups and improve feature discriminability. Second, a curriculum-aware dynamic negative sampling (curriculum-DNS) strategy adaptively filters potential false negatives and progressively emphasizes hard negatives, stabilizing convergence and enhancing representation learning. In addition, we propose a spectral–hierarchical dual-stage fusion encoder that incorporates LiDAR edge priors at shallow layers and elevation cues at deeper stages, enabling spatially aware and complementary multimodal fusion. Comprehensive experiments across three public benchmark datasets highlight the robustness and performance advantages of our network. The reference code for our work can be accessed at https://github.com/chaoreee/CLCNet**

***Index Terms*—Contrastive learning, hyperspectral image (HSI), image classification, light detection and ranging (LiDAR), self-supervised learning.**

I. Introduction

THE task of land-cover classification, as the core task of remote sensing image analysis, has emerged as a research hotspot in recent years [1], [2], [3]. Among the many remote sensing data, hyperspectral imaging, recognized by the EU as one of the 100 disruptive technologies of the future [4], provides rich and continuous spectral information, enabling the capture of subtle differences between land cover types [5], [6], [7]. Despite these advantages, classification based solely on hyperspectral image (HSI) remains challenged by interclass similarity, intraclass variance [8], limited spatial resolution [9], and noise [10]. Light detection and ranging (LiDAR), with its strong penetration capability and precise spatial and elevation data, complements the spectral richness of HSI by providing detailed spatial and structural information, making their integration essential for improving classification performance [11], [12], [13]. Consequently, many researchers have directed their attention to the combined application of HSI and LiDAR data in land-cover classification, seeking to produce more precise and dependable classification performance [14], [15], [16].

Driven by the remarkable progress of deep learning (DL) in recent years [17], DL-based methods that integrate LiDAR and HSI for joint classification have achieved significant improvements in classification accuracy [18], [19]. The main model architectures are mainly based on convolutional neural network (CNN) and transformer. CNN excels at capturing local spatial information and possesses inherent spatial prior knowledge [20]. Transformer can directly model global contextual relationships in spatial and spectral dimensions, enabling more flexible feature interaction and cross-modal alignment [21]. For example, Hong et al. [22] constructed a deep encoder–decoder framework built on CNNs to merge multimodal data and sequentially restore the inputs from fused features. Li et al. [23] utilized a CNN-driven feature reconstruction embedding method to guide fusion, integrating reconstructed HSI representations into classification features via weighted embedding. Zhao et al. [24] introduced a fusion encoder, termed cross-token attention (CTA), which leverages self-attention to achieve more effective integration of LiDAR and HSI features. Yao et al. [25] proposed a multimodal processing network built on position-shared ViT and separable convolution branches to improve the discriminative power of different modalities. Overall, these works indicate the growing effectiveness of DL architectures in advancing hyperspectral–LiDAR fusion for land-cover classification.

Moreover, a major challenge in performing integrated classification using HSI and LiDAR data is the limited availability of labeled samples. This challenge is mainly owing to the

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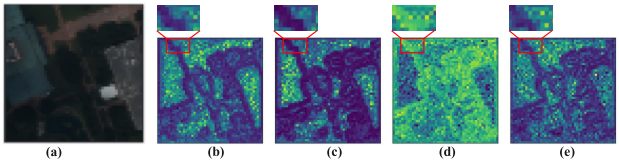


Fig. 1. Heatmap of attention to HSI-patch by different heads in MHA. (a) Original image. (b) Head 1. (c) Head 2. (d) Head 3. (e) Head 4.

high dimensionality, fine spatial details, and complex spectral characteristics of remote sensing imagery, which make accurate annotation highly dependent on specialized domain knowledge and require substantial time and labor investment [26]. Therefore, many excellent works have exploited the unlabeled sample information by introducing unsupervised learning, followed by supervised fine-tuning, which leads to a model with superior generalization ability and detection accuracy. For example, Lin et al. [27] presented an unsupervised learning scheme named spatial–spectral masked autoencoder (SS-MAE) for multimodal classification of HSI and LiDAR data. The model comprises a spatial branch that trains by filling in missing image patches and a spectral branch that reconstructs withheld spectral bands. By combining these two reconstruction tasks, the framework effectively exploits structural and spectral cues.

In these self-supervised networks, multihead attention (MHA) can simultaneously mine diverse feature information from various subspaces and scales without label guidance, thereby enhancing the richness and discriminability of feature representations. However, MHA also faces the challenge of attention redundancy, where information extracted by different heads is highly overlapping. Prior studies (e.g., Zhou et al. [28]) have indicated that standard MHA is prone to introducing redundancy due to its uniform partitioning of subspaces, which limits model performance. Meanwhile, in the hyperspectral domain, Li et al. [29] demonstrated from the perspective of band selection that the inherent structural correlation between spectral bands leads to information redundancy, hindering effective feature expression. Such feature extraction patterns tend to exacerbate the self-supervised model's reliance on irrelevant or repetitive information, thereby reducing the utilization efficiency of critical details. As shown in Fig. 1, we present the attention heatmaps of an HSI patch under four-head self-attention. It can be observed that Head 1 and Head 2 focus on highly similar regions. This phenomenon arises because, during MHA computation, different heads may assign disproportionately high attention weights to similar spectral bands. As a result, the model concentrates on redundant features, failing to fully exploit the spectral diversity inherent in the data.

As a common self-supervised framework, contrastive learning acquires feature embeddings by reducing the distance between positive pairs while enlarging the gap with negative pairs, directly optimizing the similarity between sample pairs and exhibiting strong generalization ability [30], [31]. Within the domain of remote sensing land-cover classification, these contrastive learning-based land cover classification methods [32], [33], [34] typically learn discriminative feature representations by building positive and negative sample pairs, and then transfer the pretrained features to downstream tasks to enhance generalization performance in few-shot or cross-domain scenarios. To overcome the difficulty caused by the lack of labeled data, Jia et al. [32] introduced a contrastive learning-based collaborative framework that maps multimodal features into a common latent space, enabling comprehensive exploitation of complementary information between the two types of RS data. Xu et al. [33] introduced a semantic contrastive loss by treating HSI and LiDAR feature maps from the same region as positive pairs, enabling the model to learn cross-modal complementary information from unlabeled data. Wang et al. [34], inspired by MoCo [35], adopted a neighborhood-based data augmentation strategy to replace the original positive samples with nearby samples, thereby enhancing the robustness of positive sample representations. It can thus be seen that contrastive learning methods can also provide an effective baseline for hyperspectral–LiDAR fusion classification.

However, most of these contrastive learning methods are mainly concerned with designing positive and negative sample pairs, without adequately addressing the issue that interclass similarity and intraclass heterogeneity in remote sensing images can result in inappropriate negative samples, ultimately becoming a performance bottleneck. As Tianhui Cai et al. [36] pointed out, 95% of easily distinguishable negative samples are unnecessary and insufficient, while the most challenging 0.1% are the most critical, as they may include false negatives that impair the network's discriminative ability. From what we have observed, in the multimodal fusion of HSI and LiDAR, the design of contrastive learning frameworks that incorporate negative sample selection remains largely unexplored.

In DL, curriculum learning [37], inspired by human cognitive development, has emerged as an effective training paradigm in which models learn from easy samples first and gradually transition to harder ones. This approach has been increasingly adopted to enhance training stability, accelerate convergence, and improve generalization performance. It has been extensively applied across multiple fields, including computer vision [38], [39], reinforcement learning [40], [41], and graph learning [42], [43]. For example, Huang et al. [44] introduced a contrastive loss function embedded with a curriculum strategy, where a difficulty-related parameter m modulates the contribution of each sample during training, enabling adaptive weighting of easy and hard examples. Zhuang et al. [45] introduced curriculum learning into the SimCLR framework by incorporating a regularization loss, progressively enabling the model to select samples from easy to hard, improving classification performance and accelerating convergence. Therefore, leveraging the strengths of curriculum learning in difficulty modeling and dynamic sample scheduling offers a novel and promising solution to the negative sample selection problem in contrastive learning for remote sensing. However, how to correctly assess the difficulty of samples and determine when to introduce more complex ones during training remains a question worth considering [46]. Zheng et al. [47] propose C2PNet, which introduces a curriculum-based contrastive learning strategy that dynamically categorizes con-

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sensual negative samples into easy, hard, and ultrahard levels and adaptively reweights them, enabling the model to progressively incorporate more constraining negatives from easy to hard during training, thereby stabilizing optimization and gradually tightening the solution space.

To address the two aforementioned issues.

1) *Functional Redundancy in MHA Under Self-Supervised Settings:* In multihead self-attention, heads are expected to specialize across feature subspaces. Strong interband correlations often cause head overlap in label-free self-supervision on HSI, creating redundant attention and underused features. Such redundancy biases reconstruction toward easy content and weakens contrastive separability, hurting downstream classification.
2) *Improper Negative Sample Design in Contrastive Learning:* Current multimodal fusion methods based on contrastive learning largely rely on predefined positive and negative sample pairs. However, in remote sensing scenarios, the high interclass similarity and strong intraclass heterogeneity often result in semantically similar samples being incorrectly treated as negative ones—so-called false negatives. These misleading samples can significantly disrupt the learning process. Moreover, truly hard negatives are often not emphasized adequately during training, resulting in a model that performs well on easy samples but lacks discriminative power for challenging ones.

In response to these challenges, we introduce a curriculum-learning-based contrastive learning network (CLCNet), which not only alleviates the issue of false negatives caused by high interclass similarity and intraclass heterogeneity in remote sensing data, but also progressively assigns greater weights to hard negative samples. The principal contributions of our research are listed below.

1) A spectral-hierarchical MHA mechanism is introduced, where spectral channels are reordered and grouped according to similarity. Each group is processed by an independent attention head, effectively decoupling attention responses and enhancing the discriminative capability of individual heads.
2) A curriculum-aware dynamic negative sampling (curriculum-DNS) strategy is proposed to adaptively partition negative samples in an easy-to-hard manner while filtering potential false negatives. This design stabilizes contrastive optimization, emphasizes informative hard negatives, and improves feature discriminability under limited supervision.
3) A self-supervised contrastive learning framework, termed CLCNet, is developed with a spectral-hierarchical dual-stage fusion (SHDF) encoder. LiDAR edge information is incorporated at the shallow stage as a structural prior to enhance spatial awareness, while LiDAR elevation cues are integrated at deeper stages to refine multimodal feature fusion, enabling effective complementarity between hyperspectral and LiDAR data.

The subsequent content of this article is arranged as follows. Section II will present the specific methodology adopted in this work, and Section III will describe the related datasets and experimental outcomes. Finally, some summary notes are summarized in Section IV.

II. Methodology

A. Method Overview

The primary motivation behind CLCNet is to adopt a curriculum-guided contrastive learning framework that progressively shifts from easy to hard negatives, while suppressing false negatives to guide the training of the SHDF encoder.

As illustrated in Fig. 2, the overall network pipeline is outlined as follows.

1) Data Preprocessing: HSI and LiDAR data are first divided into equal-sized patches to generate query-key views for contrastive learning. Each patch is then augmented twice using transformations such as Gaussian blur, flipping, and spectral distortions to produce the query samples $X^q_{H\&L}$ and key samples $X^k_{H\&L}$.

2) Hierarchical-Spectral Dual-Stage Fusion Encoder: To extract complementary features and achieve cross-modal fusion, the query samples $X^q_{H\&L}$ and key samples $X^k_{H\&L}$ are fed into our proposed hierarchical-spectral dual-stage fusion encoder. At the shallow stage, HSI features are fused with LiDAR edge priors and processed by the spectral-hierarchical edge attention (S-HEA) module to enhance the discriminative capacity of each attention head. The encoder further fuses the features with LiDAR elevation information at the deeper stage, enabling effective and comprehensive fusion between the HSI and LiDAR modalities. As a result, we obtain the encoded representations X^q_{encoder} and X^k_{encoder}.

3) Curriculum-Aware Dynamic Negative Sampling: The encoder features $X^q_{encoder}$ and $X^k_{encoder}$ are projected into the anchor samples q and the positive samples k_+, respectively. Following a MoCo-like approach, we maintain a dictionary queue of negative sample keys, denoted as $\{k_1, k_2, k_3, \ldots, k_n\}$. Each batch of positive samples k_+ is pushed into this queue to be part of the negative samples for the next batch, while the oldest entries are popped from the tail.

We compute the cosine similarities between q and all samples to form the set $\{S^+, S_1, S_2, \ldots, S_n\}$, where S^+ is the similarity with the positive sample, and S_i is the similarity with the i-th negative. The contrastive loss is defined as

$$\mathcal{L} = -\log \frac{\exp\left(S^+/\tau\right)}{\sum_{i=1}^{N} \exp\left(S_i/\tau\right)} \tag{1}$$

where τ is a temperature hyperparameter.

To mitigate hard and false negatives caused by high interclass similarity and intraclass variation, we adopt a curricular-DNS strategy. Details are provided in Section II-C.

B. SHDF Encoder

To extract semantic feature representations from different modalities and perform complementary feature fusion, we propose the SHDF encoder illustrated in Fig. 2. In particular, the CBAM module [48] is incorporated into the feature extraction stage to highlight informative channel-wise and spatial regions.

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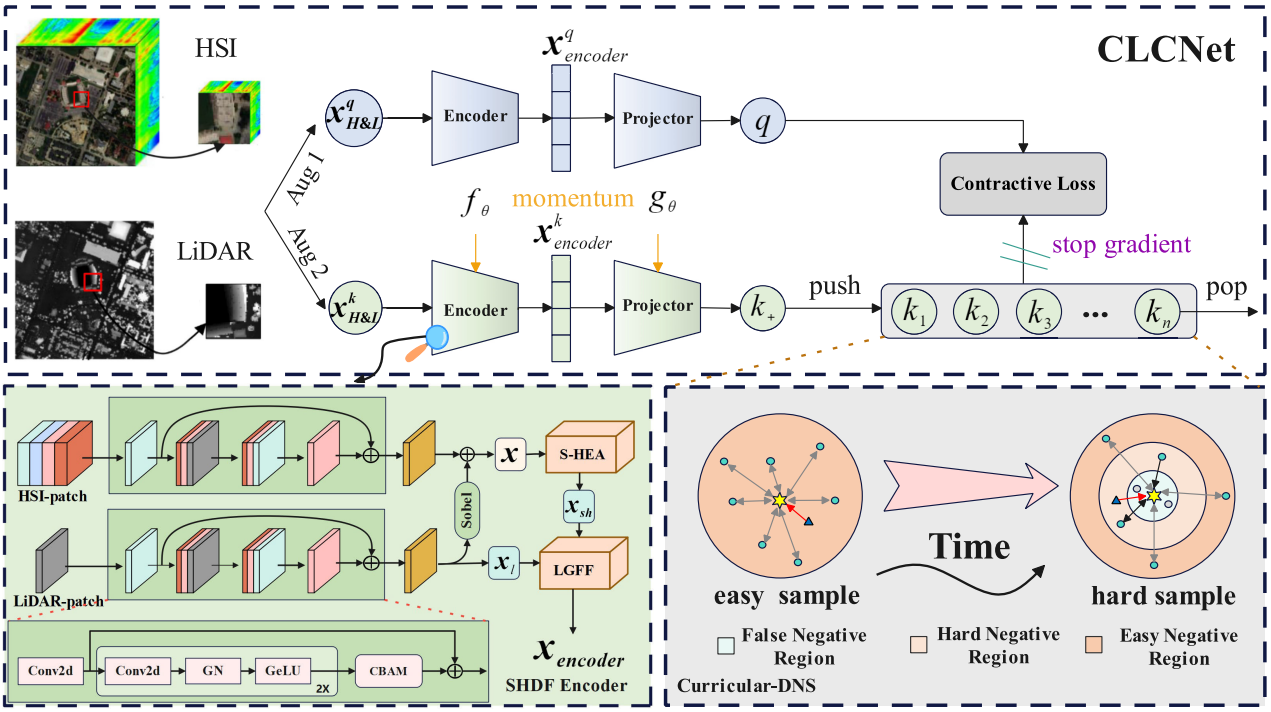


Fig. 2. Overview of the proposed self-supervised contrastive learning framework (CLCNet). Built upon the MoCo baseline, it incorporates a hierarchical-spectral dual-stage fusion encoder (SHDF encoder) and introduces a curriculum-DNS.

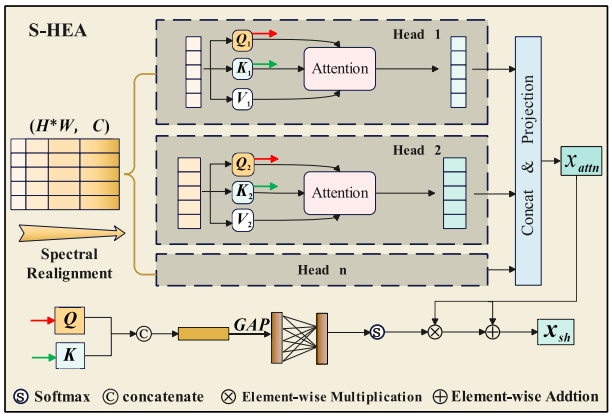


Fig. 3. Overview of the proposed S-HEA module.

Although HSI provides rich spectral information, its geometric edge representation is considerably weaker than LiDAR data. This spatial limitation can lead to blurred object boundaries and reduced spatial discriminability, particularly in the network's shallow layers.

To address this, we apply the Sobel operator early to extract edge maps from the LiDAR modality, capturing structural contours such as object boundaries and terrain transitions. These edge priors are integrated with the shallow HSI features via element-wise addition, guiding the subsequent multihead self-attention mechanism to concentrate on structurally salient regions. This edge-aware fusion enhances spatial sensitivity and facilitates more accurate modeling of object boundaries.

In the following, we briefly introduce the S-HEA module and the LiDAR-guided feature fusion module (LGFF).

1) S-HEA Module: To alleviate the redundancy in conventional MHA mechanisms, where each attention head is typically assigned to uniformly partitioned spectral subspaces in channel order without considering the semantic correlations and discriminative differences among spectral channels, we propose an S-HEA module, as illustrated in Fig. 3.

Formally, given the input spectral feature fused with LiDAR edge priors $X \in \mathbb{R}^{HW\times C}$, we first compute the pairwise interband similarity using cosine similarity

$$G_i = \frac{1}{n-1}\sum_{j\neq i}\frac{x_i^\top x_j}{|x_i|_2 \cdot |x_j|_2}, \quad x_i = X[:,i], \; x_j = X[:,j]. \tag{2}$$

Then, channels are reordered in descending order based on their similarity scores G_i, and partitioned into h semantically coherent groups: $C = [C_1, C_2, \ldots, C_h]$ with $C_1 \leq C_2 \leq, \ldots, \leq C_h$. For each group $X_i = X[:, C_i]$, we compute intragroup attention as follows:

$$\begin{aligned} H_i &= \text{Attention}(Q_i, K_i, V_i) = \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i \\ Q_i &= X_i W_Q^i, \quad K_i = X_i W_K^i, \quad V_i = X_i W_V^i \end{aligned} \tag{3}$$

where the vectors Q_i, K_i, and V_i are obtained by passing X_i through the linear projection matrices W_Q^i, W_K^i and W_V^i, respectively.

Outputs from all groups are merged and subsequently projected to derive the ultimate output

$$X_{atten} = \text{Concat}(H_1 H_2, \ldots, H_h)\, W^O \tag{4}$$

where W^o is explicitly defined as the output projection matrix that maps the concatenated MHA features back to the original feature space.

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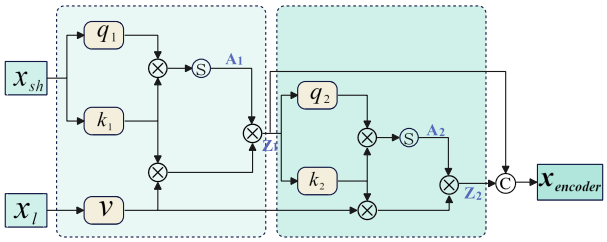


Fig. 4. Illustration of the proposed LGFF module.

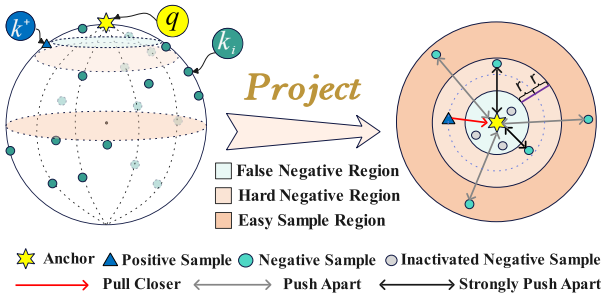


Fig. 5. Visualization of the dynamic negative sample division process on the unit hypersphere, where the feature space is divided into easy, hard, and false-negative regions to guide adaptive contrastive learning.

To enable interaction among different attention heads, we concatenate the Q and K vectors, which encode semantic attention information from different subspaces, to generate a gating weight. This gating weight is then combined with X_{attn} to adaptively adjust the output strength of each head, thereby facilitating multihead interaction and adaptive reweighting, as formulated below

$$\mathbf{w} = \sigma\left(\mathcal{F}\left(\text{GAP}\left(\text{Concat}\left(\mathbf{q}_1, \ldots, \mathbf{q}_h, \mathbf{k}_1, \ldots, \mathbf{k}_h\right)\right)\right)\right) \tag{5}$$

$$X_{\text{out}} = X_{\text{attn}} + X_{\text{attn}} \odot \mathbf{w} \tag{6}$$

where $\mathcal{F}(\cdot)$ denotes the MLP transformation, $\sigma(\cdot)$ is the sigmoid activation function, *GAP* is the global average pooling, and $\odot$ denotes element-wise multiplication. The dimensionality of $\mathbf{w}$ is now specified as a channel-wise gating vector with the same dimensionality as X_{attn}, enabling element-wise reweighting of the attention-enhanced features.

This module partitions HSI spectral channels into semantically coherent groups. It assigns each group to its own attention head, reducing redundancy, increasing interhead diversity, and improving the capture of fine-grained spectral variations.

2) LGFF Module: Although the edge priors from LiDAR help enhance the spatial boundary information in hyperspectral images (HSI), the complementary elevation information it provides remains underexploited. To address this, we design an LGFF module, as illustrated in Fig. 4.

We first compute the cross-attention between spectral and LiDAR modalities as

$$\mathbf{A}_1 = \text{Softmax}\left(\frac{q_1\left(X_{\text{out}}\right) \cdot k_1\left(X_{\text{out}}\right)^T}{\sqrt{d}}\right)$$
$$Z_1 = \mathbf{A}_1 \cdot \left(v_1\left(X_L\right) \cdot k_1\left(X_{\text{out}}\right)\right)^T \tag{7}$$

where $q_1(\cdot)$, $k_1(\cdot)$, and $v_1(\cdot)$ denote linear projections for query, key, and value. The result Z_1 represents the coarse fusion outcome, incorporating structural guidance from LiDAR.

Then, the refined fusion is performed by applying another cross-attention between Z_1 and the LiDAR features

$$\mathbf{A}_2 = \text{Softmax}\left(\frac{q_2\left(Z_1\right) \cdot k_2\left(Z_1\right)^T}{\sqrt{d}}\right)$$
$$Z_2 = \mathbf{A}_2 \cdot \left(v_2\left(X_L\right) \cdot k_2\left(Z_1\right)\right)^T. \tag{8}$$

The result Z_2 represents the fine fusion outcome, which further enhances the fusion by leveraging the semantic features of the LiDAR modality. Combining both stages produces more stable and discriminative joint feature representations.

After passing through the encoder, the HSI and LiDAR patches extract and fuse their complementary information, reducing redundant attention to the same regions. It enables the network to concentrate more on diverse regions, thereby avoiding the loss of fine details.

C. Curriculum-Aware Dynamic Negative Sampling

To map the representations from the semantic space to a space suitable for contrastive learning, X^q_{encoder} and X^k_{encoder} are passed through projection layers to generate the anchor q and the positive sample $k+$, respectively. Meanwhile, the queue $\{k_1, k_2, k_3, \ldots, k_n\}$ provides a considerable number of negative samples, among which there are many hard negatives and potential false negatives.

To mitigate the impact of hard and potentially false negatives in the sample queue, we propose curriculum-DNS that categorizes negatives into easy, hard, and potentially false types. Their training weights are progressively adjusted based on the model's learning stage.

1) Negative Sample Division: Negative samples lie on a unit hypersphere, where their cosine similarity with the anchor reflects discrimination difficulty. As shown in Fig. 5, a smaller angle implies higher similarity, indicating a harder negative.

For illustration, we project the hyperspherical feature space onto a 2D plane, where samples closer to the anchor exhibit smaller angular separations and higher similarity, making them harder to distinguish.

In early training, when the model's discriminative capacity is still weak, introducing too many hard negatives can lead to overfitting local information while neglecting global structure. Performance improvements increasingly depend on learning from these hard negatives as training progresses. Persistently focusing on easy negatives at this stage may impede further gains.

To address this, we first perform a dynamic partition of negative samples according to their similarity with the anchor and the positive pair, and then identify hard negatives and assign them higher training weights as follows:

$$N(i,t) = \begin{cases} S_i, & (1+r)\,S_i \le S^+ \\ S_i \cdot w_i, & (1+r)\,S_i > S^+ \end{cases}$$
$$\textit{where} \quad w_i = t + S_i \tag{9}$$

where S^+ denotes the cosine similarity of the positive sample pair, while s_i represents the similarity of the negative

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samples. The parameter r is a balancing hyperparameter that controls the selection range. We apply a weighting factor w_i for negative samples whose similarity falls within the selected range to emphasize them during subsequent training. The variable t will be explained in detail in Section II-C2.

Some of the hard negatives may actually be false negatives. These samples are often assigned large weights due to their high similarity to the anchor, causing the network to focus on pushing them away. However, this leads to mistakenly separating true-positive samples, resulting in negative optimization and preventing the model from converging in the correct direction. To address this issue, we further introduce a false-negative masking strategy by defining an additional partitioning interval to deactivate such samples and exclude them from the optimization. The definition is as follows:

$$S_i = \begin{cases} S_i, & (1-r)\,S_i \le S^+ \\ -\infty, & (1-r)\,S_i > S^+ \end{cases} \tag{10}$$

where the balancing hyperparameter r is used to filter out false negative samples, which are removed from the contrastive learning process and no longer participate in training.

2) Adaptive Design of Curriculum Learning: We introduce a curriculum scheduling mechanism to dynamically adjust the difficulty of negative sampling during training. In the early stages of model training, the network's discriminative ability is weak, and its ability to learn features from negative samples is poor. Directly applying the model to perform the division results in many incorrect classifications, thereby reducing the model's optimization efficiency. Therefore, we introduce an adaptive curriculum learning mechanism

$$t^{(l)} = \alpha t^{(l-1)} + (1-\alpha)\arg^{(l)}$$
$$\text{where} \quad \arg^{(l)} = \frac{1}{B}\sum_{i=1}^{B} S^{+(l)} \tag{11}$$

where $\text{avg}^{(l)}$ denotes the average similarity of B positive pairs in the l-th batch. The parameter α is a momentum term, and t is a dynamic factor that adjusts the curriculum difficulty. It is initialized to 0. As the model improves at identifying positive pairs, the average similarity increases, thereby raising the difficulty level accordingly. Overall, the above procedure forms a complete pipeline including dynamic negative partitioning, false-negative masking, and curriculum-based reweighting.

Thus, we obtain the final loss function

$$\mathcal{L} = -\log \frac{\exp\left(S^+/\tau\right)}{\sum_{i=1}^{N} \mathbb{I}_{\{S_i \neq -\infty\}} \cdot \exp\left(N\left(i,t\right)/\tau\right)} \tag{12}$$

where $N(i,t)$ is defined in (9), representing a negative sample's dynamically adjusted similarity score, based on its difficulty level and the current training stage.

Meanwhile, Algorithm 1 provides the corresponding pseudocode of the proposed pipeline, which facilitates a clearer understanding of the curriculum learning strategy adopted in this work.

Algorithm 1 Pseudocode of the Curriculum-Guided Contrastive Learning in CLCNet

```
# f_q: encoder network for query (SHDF Encoder + Projector)
# f_k: encoder network for key (momentum encoder)
# queue: dynamic negative sample dictionary
# r: hard/false negative balancing coefficient
# t: curriculum difficulty factor
f_k.param ← f_q.param # initialize momentum encoder
for (X^H, X^L) in loader do
  X_q ← Augment(X^H, X^L) # query view augmentation
  X_k ← Augment(X^H, X^L) # key view augmentation
  z_q ← f_q.forward(X_q) # query features
  z_k ← f_k.forward(X_k) # key features
  z_k ← detach(z_k) # stop gradient
  q ← normalize(z_q)
  k^+ ← normalize(z_k)
  S^+ ← q · k^+ # positive similarity
  S_i ← q · queue[i] # negative similarities
  for each S_i do
    if (1 − r)S_i > S^+ then
      S_i ← −∞ # false negative suppression
    else if (1 + r)S_i > S^+ then
      S_i ← (t + S_i) · S_i # hard negative reweighting
    end if
  end for
  Loss ← −log exp(S^+/τ) / Σ_{i=1}^{N} 𝕀_{S_i≠−∞} · exp(S_i/τ)
  # curriculum-aware contrastive loss
  Loss.backward()
  update(f_q.param) # update query encoder only
  t ← αt + (1 − α) · mean(S^+) # curriculum update
  f_k.param ← μf_k.param + (1 − μ)f_q.param
  # momentum encoder update
  enqueue(queue, k^+)
  dequeue(queue) # maintain dictionary size
end for
```

4) Gradient Analysis: To understand how our proposed method emphasizes hard negatives, we analyze the gradient of the loss function in terms of negative sample similarity. For a negative sample S_i, the gradient of the original MoCo loss is

$$\frac{\partial \mathcal{L}_{\text{MoCo}}}{\partial S_i} = \frac{1}{\tau} \cdot \frac{\exp\left(S_i/\tau\right)}{\sum_{j=1}^{N} \exp\left(S_j/\tau\right)}. \tag{13}$$

In our method, easy negatives remain unweighted, while a dynamic factor amplifies hard negatives. Specifically, for hard negatives, the gradient becomes

$$\frac{\partial \mathcal{L}_{\text{DNS}}}{\partial S_i} = \frac{1}{\tau} \cdot (t + 2S_i) \cdot \frac{\exp\left(N\left(i,t\right)/\tau\right)}{\sum_j \mathbb{I}_{\{S_j \neq -\infty\}} \cdot \exp\left(N\left(j,t\right)/\tau\right)}. \tag{14}$$

We define the relative gradient ratio as

$$r(i) = \left| \frac{\partial \mathcal{L}_{\text{DNS}}}{\partial S_i} \Big/ \frac{\partial \mathcal{L}_{\text{MoCo}}}{\partial S_i} \right|. \tag{15}$$

For easy negatives, $r(i) \approx 1$, indicating similar treatment. As training progresses and t increases, the gradients of hard

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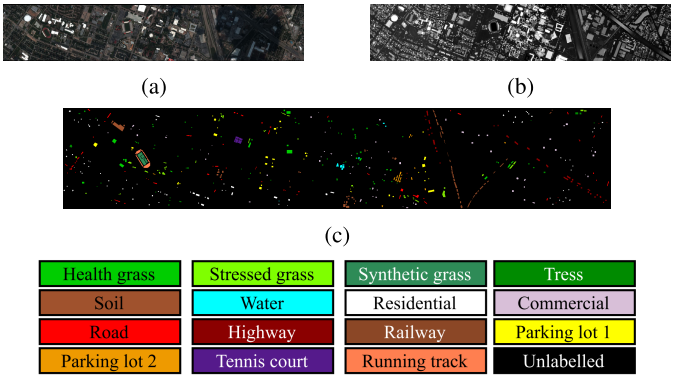


Fig. 6. Illustrations of the Houston 2013 dataset used in our experiments. From top to bottom: (a) Pseudo-color HSI image, (b) LiDAR image, and (c) ground truth labels.

negatives are progressively amplified, that is, $r(i) > 1$, guiding the model to focus more on informative but challenging samples. In contrast, false negatives are deactivated and contribute no gradient, effectively reducing optimization noise.

It is worth mentioning that after obtaining the pretrained model through contrastive learning, we added a classification head on top of the encoder for fine-tuning with supervised information, using the cross-entropy formula as the loss function shown below

$$\mathcal{L}_{\mathrm{CE}} = -\sum_{k=1}^{n} p(x_k) \log q(x_k) \tag{16}$$

where $p(x_k)$ refers to the predictive distribution over class k for a given pixel, and $q(x_k)$ denotes its corresponding expected distribution.

III. Experiments and Analysis

To evaluate the effectiveness and generalizability of our proposed method, we conducted extensive experiments on three publicly available benchmark datasets: Houston 2013, Trento, and MUUFL Gulfport. We conducted a comparison between our architecture and several advanced methods and designed various experiments to assess the superiority and feasibility of our approach comprehensively.

A. Datasets and Evaluation Indicators

In this section, we first present the three publicly available datasets through graphs and charts, followed by the evaluation indicators employed in our study.

1) Houston 2013 Dataset: The Houston 2013 dataset was released as part of the IEEE GRSS Data Fusion Contest 2013. It comprises a HSI consisting of 144 spectral channels covering the 380–1050 nm range and a co-registered single-band LiDAR image of the same region. Both modalities provide 2.5 m spatial detail across an image extent of 349 × 1905 pixels. The imagery was acquired from the University of Houston grounds as well as adjacent metropolitan regions, encompassing landscapes such as metropolitan districts, suburban zones, and farmland. In total, 15 land-cover categories are labeled, as illustrated in Fig. 6.

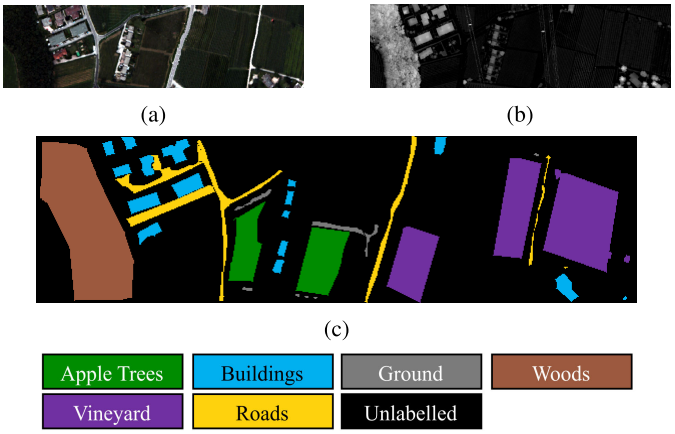


Fig. 7. Illustrations of the Trento dataset used in our experiments. From top to bottom: (a) Pseudo-color HSI image, (b) LiDAR image, and (c) ground truth labels.

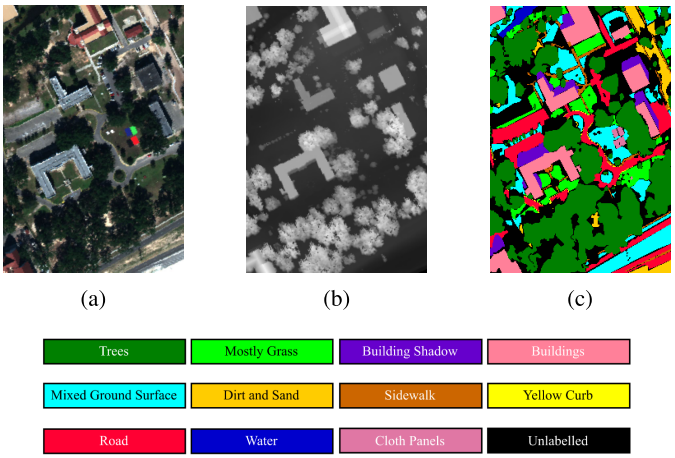


Fig. 8. Illustrations of the MUUFL dataset used in our experiments, including: (a) Pseudo-color HSI image, (b) LiDAR image, (c) ground truth labels.

2) Trento Dataset: The Trento dataset was collected in northern Italy's agricultural region near Trento. HSI data were acquired using the AISA Eagle sensor, providing 63 spectral bands spanning 402.89–989.09 nm, while a single-band LiDAR image was obtained through the Optech ALTM 3100EA system. Both data sources share a spatial resolution of 1 meter and cover an area of 600 × 166 pixels. Six land-cover categories are annotated, as shown in Fig. 7.

3) MUUFL Gulfport Dataset: The MUUFL Gulfport dataset was released through a collaboration between the University of Mississippi and the University of Southern Mississippi. It provides hyperspectral and LiDAR imagery collected over the University of Florida's Agricultural Experiment Station (MUUFL) in Gulfport, Mississippi. The hyperspectral data comprise 72 bands covering 400–1000 nm wavelengths, while the LiDAR data contain four bands. Both datasets share a spatial resolution of 1 meter with an image size of 325 × 337 pixels. Eleven land-cover categories are labeled, as illustrated in Fig. 8.

Table I reports the number of training and testing samples for each class across the three datasets.

4) Evaluation Indicators: To comprehensively assess the effectiveness of the proposed method, we evaluate the

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TABLE I
TRAIN–TEST DISTRIBUTION OF SAMPLES FOR HOUSTON, TRENTO, AND MUUFL DATASETS

NO.	Houston2013 dataset			Trento dataset			MUUFL dataset		
	Class Name	Train	Test	Class Name	Train	Test	Class Name	Train	Test
C01	Healthy Grass	198	1053	Apple Trees	20	3905	Trees	150	23096
C02	Stressed Grass	190	1064	Buildings	20	2778	Mostly Grass	150	4120
C03	Synthetic Grass	192	505	Ground	20	374	Mixed Ground Surface	150	6732
C04	Trees	188	1056	Woods	20	8969	Dirt and Sand	150	1676
C05	Soil	186	1056	Vineyard	20	10317	Road	150	6537
C06	Water	182	143	Roads	20	3052	Water	150	316
C07	Residential	196	1072				Buildings Shadow	150	2083
C08	Commercial	191	1053				Buildings	150	6090
C09	Road	193	1059				Sidewalk	150	1235
C10	Highway	191	1036				Yellow Curb	150	33
C11	Railway	181	1054				Cloth Panels	150	119
C12	Parking Lot1	192	1041						
C13	Parking Lot2	184	285						
C14	Tennis Court	181	247						
C15	Running Track	187	473						
-	Total	2832	12197	Total	819	29395	Total	1650	52037

classification results on three benchmark datasets through several well-recognized metrics, including overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (Kappa $\times 100$). Here, OA is defined as the percentage of samples that are correctly identified among the total number of test samples, which serves as a direct indicator of the overall predictive performance. AA denotes the AA computed over each class, which is particularly useful for evaluating minority categories. As for the Kappa coefficient, it evaluates the consistency between predicted and ground-truth labels while correcting for random chance, thus providing a more robust indicator under class imbalance. Combined, these metrics deliver a comprehensive assessment of model performance.

B. Classification Results and Analysis

In this section, we provide the experimental details of our specific classification task and demonstrate our network's performance on three public datasets, along with a comparative analysis with seven other advanced networks.

1) Experiment Details: All experiments were conducted on a platform equipped with an NVIDIA RTX 3090 GPU. The model was optimized using the Adam optimizer with a learning rate of 0.001. During the pretraining phase, the batch size was set to 64 and increased to 128 during the fine-tuning stage. The patch size was fixed at 11×11 throughout the entire training process.

2) Classification Results: In the experiments, we selected seven state-of-the-art networks for comparative evaluation, including S2ENet [49], MFT [50], EXViT [25], HCT [24], EndNet [22], GLTNet [51], NNCNet [34], IDNet [52], and C2PNet [47]. Specifically, Ding et al. [51] introduced a joint classification approach that integrates CNNs and transformers, utilizing multiscale feature integration and probabilistic decision strategies to combine fine-grained spatial information with broad spectral features derived from HSI and LiDAR. Fang et al. [49] introduced a spatial–spectral enhancement module (S2EM) that facilitates cross-modal interaction, where LiDAR features are utilized to strengthen the spatial representations of hyperspectral data, and hyperspectral features are employed to improve the spectral representations of LiDAR data. Roy et al. [50] introduced a multimodal fusion transformer (MFT) incorporating multihead cross-patch attention (mCrossPA), which exploits complementary modalities such as LiDAR to boost HSI classification, enhancing the generalization of the CLS token and achieving improved performance within a hierarchical feature space. Liang et al. [52] proposed IDNet to enhance hyperspectral-LiDAR fusion through the incorporation of intensity constraints and detail-enhanced representations. It should be noted that C2PNet [47] was originally proposed for image-domain tasks, where a novel curriculum contrastive regularization strategy is introduced to construct a consensual contrastive space by providing more reasonable lower bound constraints and dynamically reweighting the importance of different negative samples. We incorporate this curriculum learning strategy into our network to replace the proposed one for comparative experiments, with the aim of evaluating the applicability and classification capability of our curriculum learning idea in hyperspectral-LiDAR fusion classification.

During training, the proposed method exhibits stable and smooth convergence behavior across different datasets, without noticeable oscillations, and reaches a consistent low-loss region, indicating reliable optimization and no evident training bias.

Table II presents the classification outcomes on the Houston 2013 dataset. Our proposed CLCNet achieves state-of-the-art performance in terms of OA, AA, and the Kappa coefficient. Since our method adopts a fusion strategy that leverages the prior knowledge provided by LiDAR data, classes with distinct elevation or structural characteristics generally benefit from the complementary information. However, the elevation patterns of grass-related classes in LiDAR data are nearly indistinguishable, and their structural edge responses are relatively weak. Consequently, separating healthy grass from stressed grass

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TABLE II

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE HOUSTON 2013 DATASET

	EndNet	ExViT	GLTNet	HCT	MFT	S^2ENet	NNCNet	IDNet	C2PNet	Proposed
Healthy Grass(%)	82.53±1.23	81.29±0.06	83.10±0.16	82.24±0.98	80.53±2.58	92.97±1.20	**93.18±0.62**	81.77±0.72	82.91±2.83	78.06±3.78
Stressed Grass(%)	82.05±0.78	83.83±2.03	84.59±1.23	94.74±2.04	84.49±2.31	94.87±1.02	92.49±0.16	85.15±1.33	85.24±0.84	**98.59±1.02**
Synthetic Grass(%)	**100±0.00**	98.42±0.32	99.01±0.08	93.47±1.33	98.22±0.32	98.42±0.02	99.94±0.04	96.24±0.32	97.82±1.02	**100±0.00**
Trees(%)	91.57±1.02	92.80±1.32	91.67±2.31	97.82±0.95	89.30±3.62	96.78±1.30	96.98±1.02	90.53±1.10	92.51±1.48	**99.15±0.12**
Soil(%)	99.34±0.96	**100±0.00**	98.30±0.91	99.62±0.04	**100±0.00**	**100±0.00**	96.67±0.13	99.81±0.03	**100±0.00**	**100±0.00**
Water(%)	95.11±0.62	95.80±1.20	90.21±1.26	83.92±3.00	95.80±1.03	**100±0.00**	94.94±0.13	96.50±0.36	99.30±0.04	95.80±0.48
Residential(%)	83.12±1.36	81.53±1.58	87.78±1.58	89.37±1.66	74.35±3.66	85.91±1.30	91.98±1.85	86.47±1.54	86.19±2.56	**92.63±1.23**
Commercial(%)	85.76±2.01	93.54±1.98	81.10±2.64	93.26±2.40	86.51±2.42	88.79±2.56	77.71±3.64	92.31±0.58	**95.35±1.12**	**95.35±1.06**
Road(%)	78.47±1.46	80.08±2.61	81.16±2.31	86.97±2.20	87.72±2.00	87.06±3.01	85.36±1.72	**93.77±1.06**	86.50±2.88	91.22±2.28
Highway(%)	63.61±0.64	81.95±3.69	62.84±3.66	87.07±2.38	74.32±1.85	65.44±3.78	88.96±2.16	**89.38±1.92**	86.00±2.16	80.79±2.82
Railway(%)	87.67±2.44	97.15±0.96	84.91±2.01	92.79±1.00	**98.10±0.17**	96.49±0.94	96.19±0.44	98.01±0.17	92.60±1.05	96.87±0.54
Parking Lot1(%)	93.56±3.75	98.27±0.03	95.00±2.00	**99.52±0.06**	98.66±0.35	93.66±0.61	89.73±0.89	91.07±1.36	96.83±0.36	97.79±0.12
Parking Lot2(%)	80.70±2.06	74.04±3.88	91.23±1.57	**96.67±1.00**	90.18±1.68	85.26±1.27	96.34±0.62	88.42±1.45	90.18±2.26	82.11±2.63
Tennis Court(%)	**100±0.00**	**100±0.00**	96.36±0.76	**100±0.00**	**100±0.00**	99.98±0.01	96.81±0.18	**100±0.63**	**100±0.00**	**100±0.00**
Running Track(%)	98.31±0.85	97.67±0.17	97.46±1.25	**100±0.00**	98.94±0.05	99.37±0.02	99.50±0.12	**100±0.00**	**100±0.00**	**100±0.00**
OA(%)	86.28±0.97	89.70±1.10	87.09±1.26	92.60±0.76	88.70±1.65	90.25±1.56	91.25±0.45	91.59±0.26	91.37±0.48	**93.54±0.34**
AA(%)	88.12±0.78	90.43±0.68	88.71±2.01	92.50±0.96	90.48±0.99	91.67±2.46	92.86±0.88	92.58±0.46	92.76±0.82	**93.89±0.25**
$\kappa \times 100$	85.14±1.03	88.82±0.95	86.03±0.82	91.97±1.62	87.73±1.43	89.43±1.04	90.32±1.45	90.87±0.33	90.64±1.00	**92.98±0.28**

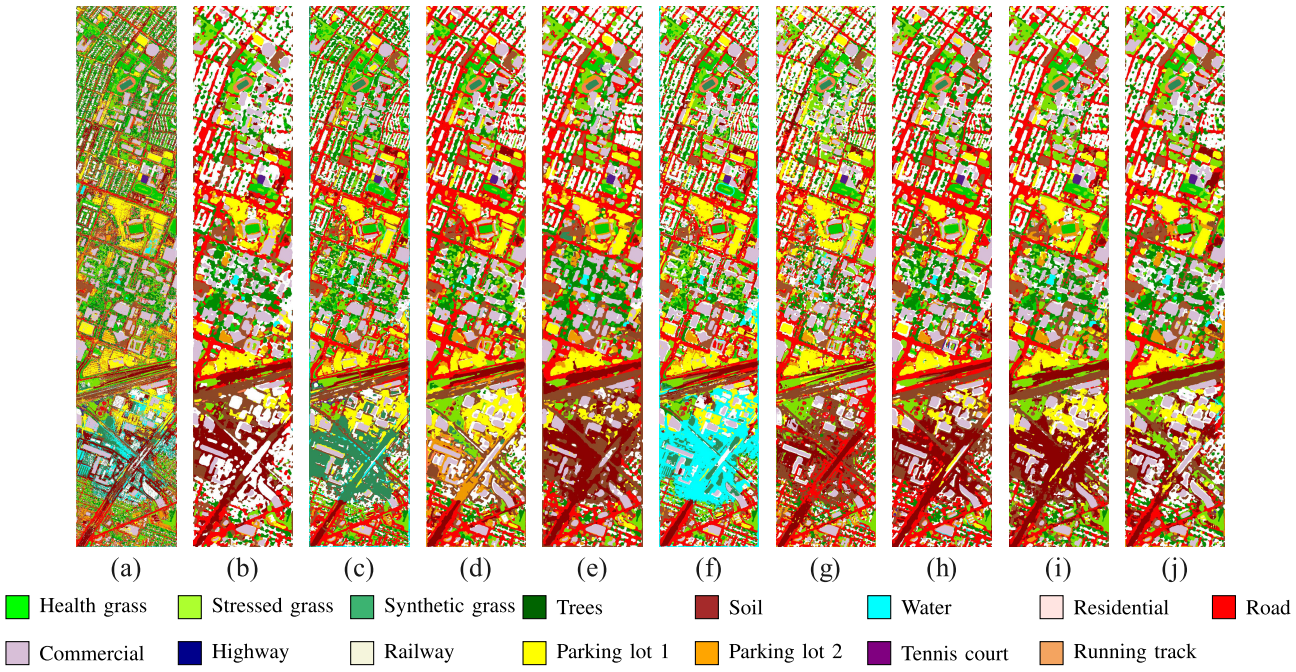


Fig. 9. Classification maps obtained by nine methods on the Houston 2013 dataset: (a) EndNet, (b) ExViT, (c) GLTNet, (d) HCT, (e) MFT, (f) S^{2}ENet, (g) NNCNet, (h) IDNet, (i) C2PNet, and (j) Proposed CLCNet.

TABLE III

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE TRENTO DATASET

	EndNet	ExViT	GLTNet	HCT	MFT	S^2ENet	NNCNet	IDNet	C2PNet	Proposed
Apple Trees(%)	74.24±2.31	96.89±0.42	97.06±0.38	98.65±0.21	95.84±0.47	99.50±0.08	99.15±0.30	**100.00±0.00**	**100.00±0.00**	98.33±0.25
Buildings(%)	94.69±0.53	97.02±0.36	99.93±0.02	99.65±0.15	99.83±0.06	98.44±0.29	94.03±1.10	**100.00±0.00**	97.14±0.16	98.30±0.35
Ground(%)	89.54±1.92	94.34±1.15	64.92±4.87	99.13±0.33	**100.00±0.00**	**100.00±0.00**	98.04±0.40	88.24±1.42	88.24±1.42	98.47±0.28
Woods(%)	97.09±0.24	99.82±0.07	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.51±0.20	**100.00±0.00**	**100.00±0.00**	99.95±0.03
Vineyard(%)	73.59±4.56	98.07±0.51	**100.00±0.00**	97.39±0.64	99.95±0.05	99.85±0.05	84.76±2.50	**100.00±0.00**	**100.00±0.00**	99.87±0.05
Roads(%)	90.81±2.41	94.99±0.59	74.48±3.12	89.76±1.94	84.43±2.18	87.51±1.37	91.92±1.80	99.02±0.22	**100.00±0.00**	93.63±0.80
OA(%)	84.85±1.27	97.96±0.48	96.39±0.62	97.79±0.51	97.78±0.44	98.42±0.29	92.98±0.60	98.43±0.08	98.14±0.18	**98.86±0.15**
AA(%)	86.66±0.73	96.85±0.66	89.40±1.35	97.43±0.58	96.68±0.71	97.55±0.46	94.57±0.75	97.88±0.32	97.56±0.17	**98.09±0.25**
$\kappa \times 100$	80.14±1.58	97.29±0.52	95.17±0.73	97.06±0.61	97.04±0.54	97.89±0.37	90.78±0.55	98.09±0.30	97.74±0.42	**98.48±0.18**

becomes more challenging, as the Sobel-enhanced LiDAR-guided fusion provides limited discriminative cues for these categories. As a result, the classification accuracy of healthy grass exhibits relatively larger fluctuation, while other classes still benefit from the complementary multimodal information. Compared to NNCNet, which adopts the same MoCo-based contrastive learning framework, CLCNet improves OA by 2.29% and achieves the highest classification accuracy across multiple categories. As shown in Fig. 9, the classification maps obtained by different methods on the Houston 2013 dataset are presented for visual comparison.

Table III summarizes the classification outcomes for the Trento dataset, and the corresponding class-level prediction maps are illustrated in Fig. 10. Due to the distinct differences

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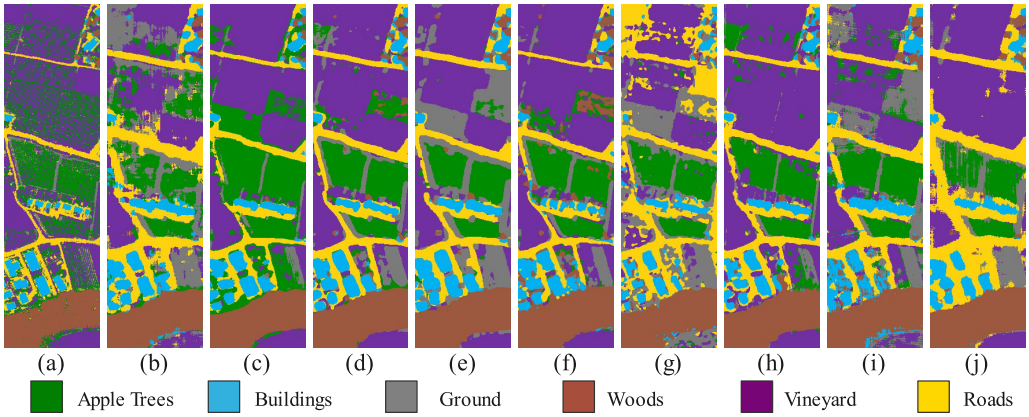


Fig. 10. Classification maps for the Trento dataset. (a) EndNet, (b) ExViT, (c) GLTNet, (d) HCT, (e) MFT, (f) S^2ENet, (g) NNCNet, (h) IDNet, (i) C2PNet, (j) Proposed CLCNet.

TABLE IV
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE MUUFL DATASET

	EndNet	ExViT	GLTNet	HCT	MFT	S^2ENet	NNCNet	IDNet	C2PNet	Proposed
Trees(%)	88.21±1.25	91.41±0.94	**94.52±1.04**	92.47±1.21	89.00±1.00	91.77±1.20	92.50±1.14	94.41±1.36	90.34±1.12	91.46±1.48
Mostly Grass(%)	87.21±1.05	83.81±2.18	86.43±1.65	83.52±2.06	84.44±1.44	86.41±1.14	86.43±2.08	88.01±1.67	89.47±1.96	**89.81±1.32**
Mixed Ground Surface(%)	70.00±3.26	77.29±3.62	72.88±2.78	83.59±1.36	77.97±3.09	**87.55±1.25**	79.13±3.14	83.84±2.21	80.33±2.18	87.21±2.28
Dirt and Sand(%)	91.35±2.71	95.23±1.23	**98.75±0.96**	93.85±1.44	96.18±0.98	95.41±0.41	95.05±0.99	97.91±0.25	96.06±0.98	97.14±0.42
Road(%)	91.27±1.03	83.97±2.97	86.48±1.46	89.72±1.27	90.55±2.00	85.91±0.19	90.44±1.13	**91.57±1.65**	90.41±2.12	91.19±1.78
Water(%)	98.42±0.63	**100±0.00**	**100±0.00**	99.05±0.25	**100±0.00**	99.68±0.03	**100±0.00**	**100±0.00**	99.68±0.12	96.20±0.36
Building Shadow(%)	87.66±1.66	89.77±2.17	93.42±0.93	90.69±1.39	91.69±0.98	**94.14±0.41**	93.13±1.47	92.32±1.12	92.70±0.63	91.74±1.05
Building(%)	92.51±1.20	96.57±1.20	**97.14±1.41**	94.73±1.37	92.99±1.32	95.65±0.65	96.32±1.75	96.57±1.20	94.33±1.04	96.67±0.46
Sidewalk(%)	77.57±3.75	67.21±2.58	60.65±3.56	72.96±1.53	66.64±4.08	**78.14±2.41**	76.60±3.20	63.81±4.71	74.17±1.21	61.13±4.68
Yellow Curb(%)	**100±0.00**	87.88±2.81	81.26±1.06	78.79±2.09	66.67±3.66	84.85±1.58	87.88±2.81	81.82±0.92	93.94±0.40	96.97±0.24
Cloth Panels(%)	**100±0.00**	99.16±0.60	99.16±0.62	99.16±0.62	99.16±0.62	99.16±0.34	99.16±0.44	99.16±0.16	99.16±0.16	99.16±0.62
OA(%)	86.59±1.03	88.20±1.20	89.70±0.12	90.09±0.25	87.76±0.20	90.47±0.37	90.27±0.44	90.60±1.04	89.43±0.88	**90.88±0.16**
AA(%)	89.47±0.80	88.39±1.09	88.29±0.52	88.96±0.29	86.84±0.42	**90.79±1.93**	90.60±1.18	89.04±1.29	90.69±0.23	**90.79±0.21**
$\kappa \times 100$	82.50±1.40	84.49±1.28	86.40±0.14	86.96±0.33	84.03±1.71	87.48±0.25	87.20±0.24	87.55±0.88	86.17±0.72	**88.01±0.24**

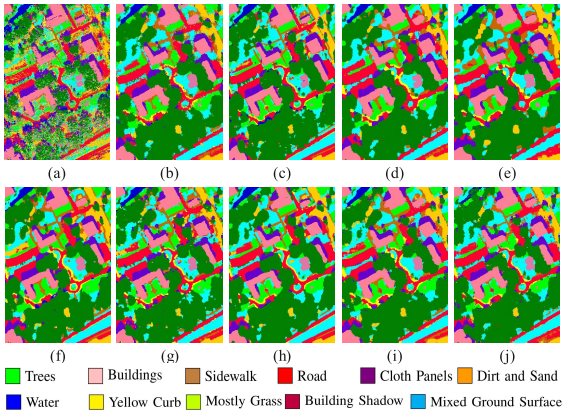


Fig. 11. Classification maps obtained by nine methods on the MUUFL dataset: (a) EndNet, (b) ExViT, (c) GLTNet, (d) HCT, (e) MFT, (f) S^2ENet, (g) NNCNet, (h) IDNet, (i) C2PNet, and (j) Proposed CLCNet.

between classes in the Trento dataset, we limited the amount of training data, using only a few samples per class. Even under this limited-data setting, our proposed CLCNet maintains high performance in overall OA, AA, and the Kappa coefficient. In the visualized results, CLCNet demonstrates a better balance and tradeoff across different categories, exhibiting stronger overall consistency in the classification map. Although CLCNet does not achieve the highest accuracy in every class, it consistently delivers competitive results across all categories. This balanced performance leads to the best OA, AA, and Kappa, highlighting the robustness and generalization ability of the proposed method.

TABLE V
MODEL COMPLEXITY COMPARISON

Method	EndNet	ExViT	GLTNet	HCT	MFT	S2ENet	NNCNet	IDNet	C2PNet	CLCNet
#Param (K)	91.97	229.10	545.63	728.35	313.07	270.87	387.4	148.04	114.61	114.61
MFLOPs	0.09	366.74	312.28	71.20	202.44	262.13	78.10	67.86	31.12	31.12

Table IV presents the classification results on the MUUFL dataset, with Fig. 11 showing the category-wise prediction maps. Compared with other strong baselines, our proposed CLCNet achieves the best overall performance again. CLCNet demonstrates more precise boundary delineation and superior overall classification performance across most categories in the visual results. However, the model performs relatively poorly on the "Sidewalk" class. It is likely due to the narrow, strip-like spatial structure of sidewalks, which makes them challenging to represent when using fixed-size patches as the basic sample units. In particular, under the contrastive learning framework, data augmentation may further obscure this class's already weak boundary features, leading the model to misclassify it as similar neighboring classes (e.g., background or adjacent surfaces), thus resulting in classification errors.

Our proposed CLCNet achieves outstanding and well-balanced performance across the Houston 2013, Trento, and MUUFL datasets, with particularly notable results on Houston 2013. It can be attributed to two main reasons: first,

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TABLE VI
ABLATION STUDY OF DIFFERENT COMPONENTS

Cases	S-HEA	Sobel-Prior	LGFF	Pretrain	Curriculum Learning	OA (%)	AA (%)	$\kappa \times 100$
1	-	✓	✓	-	-	90.27	91.25	89.43
2	✓	-	✓	-	-	90.15	91.44	89.31
3	✓	✓	-	-	-	89.72	91.20	88.83
4	✓	✓	✓	-	-	91.23	92.43	90.39
5	✓	✓	✓	✓	-	92.18	93.09	91.51
6	✓	✓	✓	✓	✓	**93.54**	**93.89**	**92.98**

the higher image resolution of the Houston 2013 dataset provides a broader and more informative spatial context for contrastive learning, offering richer reference samples; second, the curriculum learning strategy enables a smoother transition from easy to hard negative samples, which helps reduce the likelihood of misclassification. In addition, we further compare the computational time and parameter complexity of different competing methods, as summarized in Table V.

C. Ablation Study

To further demonstrate the effectiveness of the proposed CLCNet, we performed a set of ablation studies. These include ablations on different components of the SHDF encoder, analysis of the impact of the proposed curriculum learning strategy, and a visual comparison between the spectral-hierarchical MHA mechanism and the standard MHA mechanism.

1) Effectiveness of Different Components in the SHDF Encoder: We begin by examining the effectiveness of each component within the SHDF encoder, including Sobel Prior, S-HEA, and LGFF, under a setting without pretraining. As shown in the ablation results in Table VI, the strong performance of the encoder is clearly attributed to the combined contributions of these components. When the Sobel prior is completely removed, the OA decreases by 1.08%, indicating that the structural guidance provided by Sobel-enhanced LiDAR features plays a critical role in the proposed framework. When the raw LiDAR data is used as the prior instead of Sobel edges, the OA further decreases by 0.42%, suggesting that directly using smooth elevation information without explicit structural enhancement may provide less effective structural guidance for attention-based multimodal fusion. In contrast, the Sobel operator explicitly emphasizes local high-frequency structural cues, which are more compatible with attention-based feature modeling and enable the network to focus on semantically informative spatial regions.

2) Effectiveness of the Curriculum Learning Strategy: As shown in Table VI, at the quantitative level, the curriculum learning strategy outperforms the direct pretraining approach by 1.36% in terms of OA. To further verify the effectiveness of the easy-to-hard training paradigm, we visualize the pretrained features using t-SNE under both settings, as illustrated in Fig. 12. From left to right, the subfigures represent: pretrained features without curriculum learning, with curriculum learning, and with curriculum learning followed by fine-tuning. The star-shaped markers denote the class centers. It can be noted that without curriculum learning, the feature distribution is relatively chaotic, with blurred class boundaries. In contrast, the features obtained through curriculum learning exhibit a more structured distribution with clearer interclass separability. This indicates that curriculum learning effectively guides the model to capture more discriminative feature representations during pretraining, laying a better foundation for subsequent training.

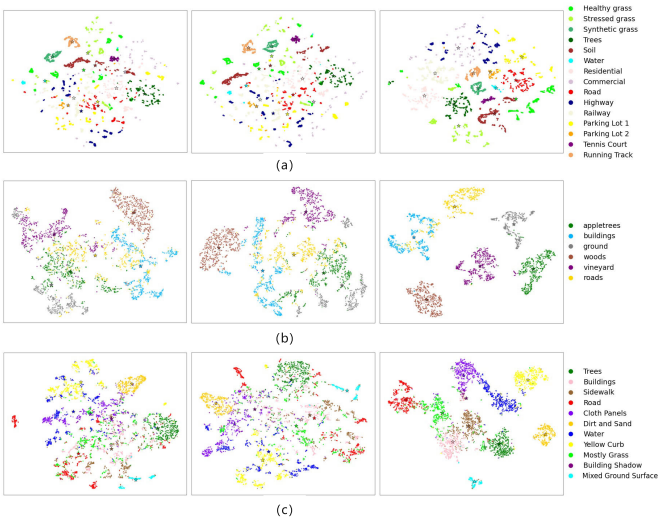


Fig. 12. Feature distributions of samples from three datasets. From top to bottom: Houston 2013, Trento, MUUFL. From left to right: pretrain without curricular learning, pretrain with curricular learning, and final feature. (a) Houston 2013, (b) Trento, and (c) MUUFL.

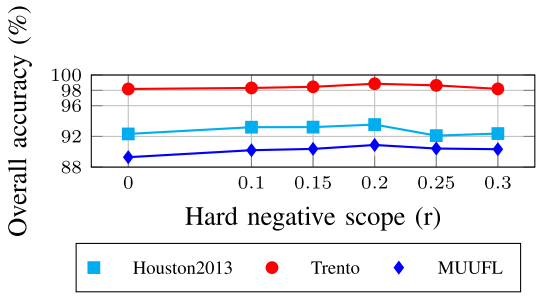


Fig. 13. Effect of hard negative scope parameter (r) on OA.

Under small-sample conditions, curriculum-guided contrastive learning facilitates more stable feature representation by gradually introducing harder samples, resulting in a more structured feature space with improved class separation.

Furthermore, we conducted a parameter sensitivity study to investigate the optimal scope of hard sample selection. As illustrated in Fig. 13, with the selection ratio r increasing, all three datasets' OA exhibits an initial upward trend. It peaks at $r = 0.2$, where the model performs best on the Trento, Houston2013, and MUUFL datasets, with OA values of 98.9%, 93.4%, and 90.9%, respectively. This observation indicates that an appropriate proportion of hard samples can effectively enhance the model's ability to learn decision boundaries, thereby improving its generalization capability. However, when r exceeds 0.2, the OA begins to decline, which may be attributed to the excessive inclusion of hard samples disrupting the training process and potentially leading to overfitting or training bias. Therefore, we consider $r = 0.2$ a relatively optimal selection ratio, striking a balance between performance enhancement and training stability.

3) Effectiveness of the Spectral-Hierarchical MHA Mechanism: To highlight the strengths of the proposed spectral-hierarchical MHA (SH-MHA) mechanism, we

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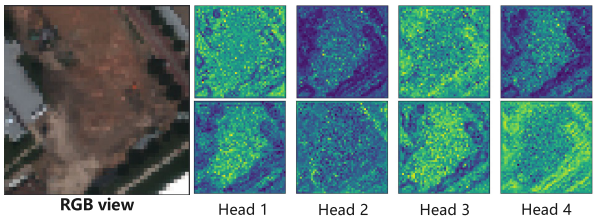


Fig. 14. Comparison of attention heatmaps from different heads in conventional MHA (top row) and the proposed spectral-hierarchical MHA (SH-MHA, bottom row), with RGB reference on the left.

visualize and compare the spatial attention distributions across different heads in Fig. 14. For ease of visualization, we extract a 50×50 patch from the input image to display the MHA maps. Specifically, the top row shows the attention heatmaps from four representative heads in a conventional MHA mechanism. In contrast, the bottom row illustrates the results from the proposed module.

As shown in Fig. 14, in the traditional MHA (top row), most attention heads focus on overlapping or redundant regions, leading to inefficient attention capacity and limited diversity in spatial feature representation. In contrast, the proposed SH-MHA (bottom row) exhibits more diverse and complementary attention patterns among different heads. Each head attends to distinct regions, indicating better head-wise disentanglement and division of labor.

This improved attention allocation enables the model to focus on more informative and diverse regions, enhancing multiscale feature extraction and overall performance on downstream tasks. The visualizations confirm the effectiveness of SH-MHA in promoting spatial diversity and strengthening attention representation.

IV. Conclusion

This article proposes a contrastive learning framework integrated with a curriculum learning strategy, named CLCNet, aimed at the combined classification of hyperspectral and LiDAR data. By introducing a spectral–hierarchical MHA mechanism, the model effectively reorders and groups spectral channels to achieve attention decoupling and enhance feature discriminability. In addition, the proposed curriculum-DNS strategy adaptively organizes negative samples and progressively emphasizes informative hard examples, leading to more stable convergence and improved representation quality. Although the framework ensures strong overall performance, the classification accuracy for individual categories still leaves considerable room for improvement. In future work, we plan to extend this study by incorporating multimodal large language models (MLLMs) in remote sensing, with the goal of enhancing category-specific representations and further addressing this limitation.

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